

数据采集-模型训练-任务评测SOP



<https://github.com/emia7/openpi>

GitHub - emia7/openpi

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数据采集 小龙 单小龙

遥控数据采集流程

- 【4090台式机】进入 `realRL` 文件夹，git branch调整为 `dev-qiuyi`
- 确认 `/home/ubuntu/realRL/experiments/souhu_franka_base/config.py` 文件的配置
 - `RESET_JOINTS` 不同任务初始pose不一样
 - `IMAGE_SIZE` = (224, 224) 固定下来了
 - `GRIPPER_SLEEP` = 0.2 固定下来了
- 启动robot server，按命令行指示选 `N`

terminal 1

```
1 source .venv/bin/activate
2 bash ./run_server.sh # 启动realRL的robot server
3
4 # 看到如下输出即为启动成功
5 * Serving Flask app 'franka_server'
6 * Debug mode: off
```

- 用3D鼠标进行采集，注意每个子任务成功后都要按 `b` 键，每个任务最后需要将机械臂移动到接近reset的位置

terminal 2

```
1 source .venv/bin/activate
2 CYCLIC=false ./scripts/collect_data.sh {task_name} {采集条数} #3D鼠标采集操作数据
3
4 # 可选task_name及对应的子任务
5 {
6     "pick-place-cup": [
```

```

7         "pick up the purple cup on the desk",
8         "approach to the blue plate and place the purple cup onto it"
9     ],
10    "stock-shelves": [
11        "pick up the stock on the desk",
12        "place the stock at the middle of the shelf"
13    ],
14    "heat-food": [
15        "open the oven door",
16        "pick up the food on the desk",
17        "place it into the oven",
18        "close the oven door"
19    ]
20 }

```

- 采集完成后处理数据，默认会存到 `realRL/ready_to_upload` 文件夹

```

realRL

1  ./scripts/process_data_local.sh {task_name} #将数据转成mp4+json+parquet格式
2
3  # 压缩成{task_name_02xx_xxepisodes}.zip文件
4
5  scp ready_to_upload/{task_name_02xx}.zip eva14:/share/guqiuyi-
   local/franka_ready_to_upload # 上传服务器，同时传到数据备份文档
6

```

无本体数据采集流程

- 【笔记本】配环境
 - 参考 [FastUMI Pro安装使用](#)，安装硬件sdk，安装monitor
- （选用）【罗姆楼新主机docker】
 - 初次开机启动docker，并打开转发

```

host terminal

1  # 仅初次开机需要
2  docker start data_collection
3  xhost +

```

- 在终端进入docker或使用vscode连接docker

host terminal

```
1 docker exec -it data_collection /bin/bash
2 # 或打开vocode通过Dev Containers插件进入docker
```

- 在docker内启动相机

docker terminal1

```
1 roslaunch xv_sdk xv_sdk.launch # clamp data启动发布和可视化已写入launch文件
```

- 采集

docker terminal3

```
1 cd ~/data_collection_ws/
```

- 采数据，参考文档对应位置，开三个终端

Terminal 1

```
1 cd ~/catkin_ws/
2 roslaunch xv_sdk xv_sdk.launch
```

Terminal 2

```
1 # conda activate fastumi 不需要
2
3 cd ~/CSW/FastUMI_Monitor_Tool/
4
5 bash fastumi_monitor_menu.sh
6
7 20
```

先在teminal 2 中打开21，再采

Terminal 3

```
1 cd ~/CSW/data
2 # 注意序列号
3 rosbag record \
4 /xv_sdk/250801DR48FP25002268/clamp/Data \
5 /xv_sdk/250801DR48FP25002268/color_camera/image \
```

采集过程中注意：clamp必须有输出，可视化界面上方中间必须有画面

- 数据量很大，笔记本一次可能采不了多少

解决方案一：采一部分，然后存到硬盘，全部采完一起转mp4+json

解决方案二：采一部分，转成MP4+json，之后继续采，转mp4+json时修改start_idx即可

- 数据格式转换

批量处理，同时会可视化出state

代码块

```
1 bash ~/CSW/umi/stage_1_vis.bash
```

- 数据筛选：检查一下mp4和json有没有不对劲的，比如pose没录到或者全0，或者是某些数据明显不对，删除这些数据，重新整理这个文件夹，后续处理需要在服务器上进行

代码块

```
1 # 先预览一下
2 python .\openpi\umi\json_sort.py --dir
  .\batch_pickup_cubes_0125\batch_pickup_cubes_0125\
3
4 # 没问题就 --apply
5 python .\openpi\umi\json_sort.py --dir
  .\batch_pickup_cubes_0125\batch_pickup_cubes_0125\ --apply
```

- 处理完成后上传服务器和数据备份文档

realRL

```
1 # 压缩成{task_name_02xx}.zip文件
2
3 scp ready_to_upload/{task_name}.zip eva14:/share/guqiuyi-
  local/umi_ready_to_upload # 上传服务器，同时传到数据备份文档
```

无本体+第三视角采集流程

- 环境配置

参考 [FastUMI Pro安装使用](#)，安装硬件sdk，安装monitor

- 数据采集

参考文档对应位置，开三个终端

代码块

```
1 cd ~/catkin_ws/  
2 roslaunch xv_sdk xv_sdk.launch
```

代码块

```
1 cd ~/catkin_ws/  
2 roslaunch xv_sdk rs_camera.launch
```

代码块

```
1 cd ~/catkin_ws/src/xv_sdk/scripts  
2 ./record.sh <output-path> <task-name>
```

- 数据格式转换

解析bag为图像

代码块

```
1 cd ~/catkin_ws/src/xv_sdk/scripts  
2 python3 extract_images_from_bag.py --bag_dir <path-to-rosbags> --topics  
  <topic1, topic2, ... >
```

将双目处理为深度  罗兆亿

将深度对齐到rgb图像

代码块

```
1 cd ~/catkin_ws/src/xv_sdk/scripts  
2 python3 align_depth.py --input_dir <path-to-images> --output_dir <path-to-  
  output-path>
```

数据备份文档

[📖 数据备份](#)

遥操数据云端解析

eval14服务器

```
1  # 解压缩
2  unzip {task-name_02xx_xxepisodes}.zip
3
4  # 从上传的多组franka遥操数据中均匀采样num_episodes条数据，生成lerobot格式A的数据集
5  export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot
6  python convert_franka_data_to_lerobot.py \
7  --input_dirs=/share/guqiuyi-
  local/franka_ready_to_upload/{task_name_02xx_xxepisodes} \
8  --input_dirs=/share/guqiuyi-
  local/franka_ready_to_upload/{task_name_02xx_xxepisodes} \
9  --repo_id=local/{franka_task_name_02xx_evalx00} \
10 --num_episodes={x0} \
11 --seed=42
```

无本体数据云端解析

eval14服务器

```
1  # 解压缩
2  unzip {umi_task_name_02xx_xxepisodes}.zip
3
4  # 从上传的多组umi数据中均匀采样num_episodes条数据，生成lerobot格式B的数据集
5  export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot
6  python convert_umi_data_to_lerobot_sampling.py \
7  --input_dirs /share/guqiuyi-
  local/umi_ready_to_upload/{umi_task_name_02xx_xxepisodes} /share/guqiuyi-
  local/umi_ready_to_upload/{umi_task_name_02xx_xxepisodes} \
8  --repo_id local/{umi_task_name_02xx_evalx00} \
9  --target_fps {10} \
10 --num_episodes {x00} \
11 --seed 42
```

模型微调

- 改 `src/openpi/training/config.py` 中的 `repo_id="local/{task_name}"`

eval14服务器

```

1 # LeRobot本地数据根目录
2 export HF_LEROBOT_HOME=/share/guqiuyi-local/huggingface/lerobot
3 export HF_DATASETS_OFFLINE=1
4
5 # 归一化, 如果报错先rm -rf ~/.cache/huggingface/datasets清除缓存
6 CUDA_VISIBLE_DEVICES=4 uv run scripts/compute_norm_stats_{遥操需要加
  skip_action10}.py --config-name pi05_{franka/umi}_finetune
7
8 CUDA_VISIBLE_DEVICES=4 XLA_PYTHON_CLIENT_MEM_FRACTION=0.9 uv run
  scripts/train.py pi05_{franka_rel6d}_finetune --exp-
  name=ckpt_{franka_pick_place_cup_0202_eval100} --overwrite # 开始训练

```

任务评测 赵大为

任务评测流程

- 【4090台式机】打开 `realRL` 和 `qiuyi/openpi` 文件夹，git branch都调整为 `dev-qiuyi`
- 复制服务器上的checkpoint到4090台式机

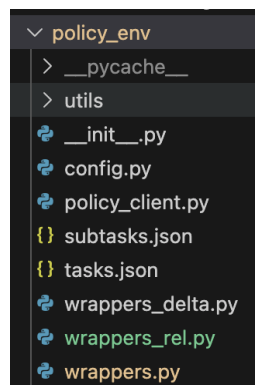
4090台式机

```

1 scp -r eval4:/share/guqiuyi-
  local/checkpoints/pi05_{franka/umi}_finetune/{ckpt_name}/19999
  /home/ubuntu/qiuyi/ckpts/{ckpt_name} # 将模型复制到本地4090

```

- 点开 `/home/ubuntu/realRL/policy_env` 目录，选择wrapper
 - 如果是跑的pi05_umi_finetune，不用改
 - 如果是跑的pi05_{franka_xxx}_finetune
 - 将wrapper.py改成wrapper_umi.py
 - 将wrapper_rel.py改成wrapper.py
- 开始评测，每测完一次后kill掉eval_policy.sh重新跑，结果记录在 [任务评测表](#)



terminal 1 realRL目录

```

1 bash ./run_server.sh # 启动realRL的robot server

```

```
1 rm -rf /home/ubuntu/qiuyi/realRL
2
3 # 无本体模型
4 uv run scripts/serve_policy_csw.py policy:checkpoint --
  policy.config=pi05_umi_rel6d_image_only_finetune --
  policy.dir=/home/ubuntu/qiuyi/ckpts/{checkpoint名称}
5
6 # 遥操模型
7 uv run scripts/serve_policy.py policy:checkpoint --
  policy.config=pi05_franka_rel6d_finetune --
  policy.dir=/home/ubuntu/qiuyi/ckpts/ckpt_franka_stock_shelves_0202_eval100/1
  9999
8
```

terminal3 realRL目录

```
1 CYCLIC=false ./scripts/eval_policy.sh {task_name} 1 # 部署测试
```

评测结果记录文档

[📄 任务评测表](#)